

Indian Statistical Institute
First Semester Examinations: 2024-25

Course Name: M. Math, 1st year
Subject Name : Topology I
Maximum Marks: 60, Duration: Three hours
Date: 20.11.2024, 2:30 PM-5:30 PM

- Answer as many questions as you can.
- Maximum marks is 60.
- You may use any results proved in class. Any other results (including those in homework problem sets) require proof.

1. a) With respect to the usual inclusion $[z_0, \dots, z_{n-1}] \mapsto [z_0, \dots, z_{n-1}, 0]$ of $\mathbb{C}P^{n-1}$ in $\mathbb{C}P^n$, show that the quotient space $\mathbb{C}P^n / \mathbb{C}P^{n-1}$ is homeomorphic to S^{2n} . 5
b) Show that there is an open set of $\mathbb{C}P^n$ containing $\mathbb{C}P^{n-1}$ which deformation retracts to $\mathbb{C}P^{n-1}$. 5

2. Let $A \in SO(2)$ be the rotation matrix which rotates a vector by angle $2\pi/5$ in the anti-clockwise direction. Prove that $D^2/[x \sim Ax]$ is homeomorphic to D^2 . 5

3. Consider $S^2 = \mathbb{C} \cup \{\infty\}$ the one point compactification of $\mathbb{C}$, and the action map $\phi : S^1 \times S^2 \rightarrow S^2$ given by

$$\phi(\omega, z) = \begin{cases} \omega z & \text{if } z \in \mathbb{C}, \omega \in S^1 \\ \infty & \text{if } z = \infty. \end{cases}$$

- a) Show that the action map ϕ is continuous. 4
b) Prove that orbit space S^2/S^1 is homeomorphic to the closed interval $[0, 1]$. 6

4. Let $A \in GL_n \mathbb{R}$.

- a) Prove that the induced map $\lambda(A)$ from $\mathbb{R}^n$ to $\mathbb{R}^n$ given by $\lambda(A)(v) = Av$ corresponding to the associated linear transformation is a proper map. 5
b) Prove that $\lambda(A)^+ : S^n \rightarrow S^n$ (the map induced on one-point compactifications) is homotopic to identity if $\det(A) > 0$. 5

5. Suppose X, Y are two simply connected spaces. Let $x_0 \in X, y_0 \in Y$ be such that there is a contractible neighbourhood containing x_0 in X (respectively, y_0 in Y). Construct the space $X \vee Y := (X \sqcup Y)/x_0 = y_0$. Prove that $X \vee Y$ is simply connected. 10

6. Show that if B is simply connected, then any covering space $p : E \rightarrow B$ for which E is path connected is a homeomorphism. 10

7. Let p, q, r be three distinct points of S^2 . Compute $\pi_1(S^2 \setminus \{p, q, r\})$. 10